

1. This scheduler ponders whether there is enough memory for the processes already admitted in the system.
medium-term scheduler
2. Match the queues to the processes in them
job queue - processes just admitted in the system
ready queue - processes waiting only on the CPU to execute
waiting/blocked queue - processes that just initiated an I/O operation
suspended ready queue - processes waiting on the CPU and memory to execute
suspended waiting/blocked queue - processes waiting on memory and I/O operations to complete
3. This scheduler ponders whether resources are sufficient to admit a new process.
long-term scheduler
4. The long-term scheduler moves processes between the _____ queue and the _____ queue.
job
ready
5. Check all that apply to CPU-bound processes
have some very long CPU bursts
use very few I/O operations
6. This scheduler is involved when a process initiates an I/O operation.
None of these answers
7. Suppose that no process is in the ready queue and the CPU is available. From which queue should the CPU scheduler select the next process to run/execute?
None of these answers
8. This scheduler controls the degree of multiprogramming.
long-term scheduler
9. The short-term scheduler moves processes between the _____ queue and the _____ queue.
On-CPU (running)
ready
10. This scheduler is involved when a process must be swapped out from the main memory.
medium-term scheduler
11. Select the best characterization of a CPU-bound process.
A process with long and frequent CPU bursts.
12. These processes are CPU-bound.
Performing a Fast Fourier transform
Performing a large matrix multiplication
13. For real time systems, these metrics are paramount objectives of a CPU scheduling policy.
fairness
meeting deadlines
14. A process P arrived at Time 0.2s. We assume that the system has plenty of memory. The

process P spent 200 ms computing and 800 ms in I/O operations. Process P completed at Time 2.4s. The waiting time for Process P is _____ s.

1.2

15. In general, a CPU scheduling policy aims to minimize these metrics.

Response time

Turnaround time

16. For interactive systems, these metrics are paramount objectives of a CPU scheduling policy.

Fairness

Balance

17. During an observation cycle, a system executed:

1) 20 processes that took 20ms each

2) 40 processes that took 30 ms each

3) 10 processes that took 40 ms each

The throughput of the system was _____ jobs per second

35

18. A process P arrived at Time 0.2s. We assume that the system has plenty of memory. The process left the ready queue the first time at Time 0.8s and completed at Time 2.4s. The response time for Process P is _____.

0.6

19. These processes are IO-bound.

Editing a document using a word processor

displaying a large matrix

loading the Fast Fourier transform of an image

downloading an app

20. For batch systems, these metrics are paramount objectives of a CPU scheduling policy.

turnaround time

CPU utilization

fairness

throughput

21. In the context of process management, match the schedulers.

Admission/Job scheduler - Long-term scheduler

Memory scheduler - medium-term scheduler

CPU scheduler - short-term scheduler